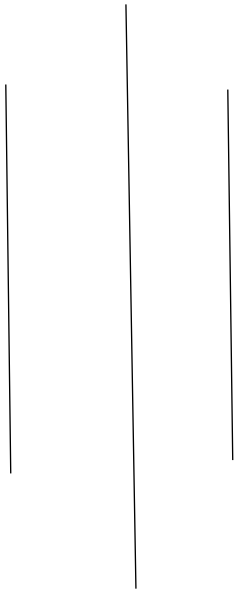
LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara, Nepal



Lab Assignment

Of

"DSA (Data Structures & Algorithm)"

Submitted By

Divya Sharma

PU Registration no: 2025-1-53-0323

Semester: 3rd

Program: BCA

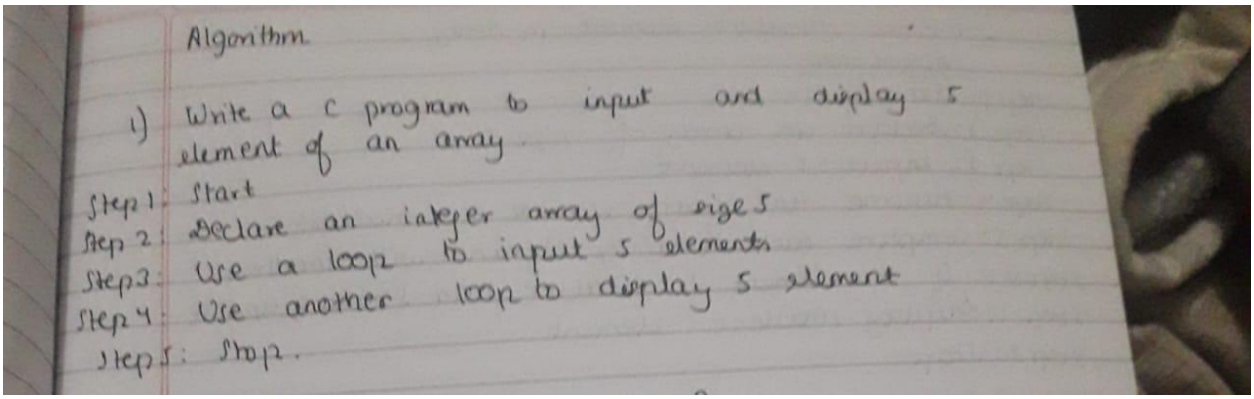
Submitted To

Ashwin Poudel

LAB:1

Q1. Program to Input and Display 5 Elements of an Array

Algorithm:



Code:

```c
#include <stdio.h>
int main()
{
    int a[5], i;
    printf("Name: Divya Sharma\n");
    printf("Registration No: 2025-1-53-0323\n");
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("The elements are:\n");
    for(i = 0; i < 5; i++)
    {
        printf("%d ", a[i]);
    }
    return 0;
}
```

Output:

```
Name: Divya Sharma
Registration No: 2025-1-53-0323

Enter 5 elements:
3
3
4
5
6

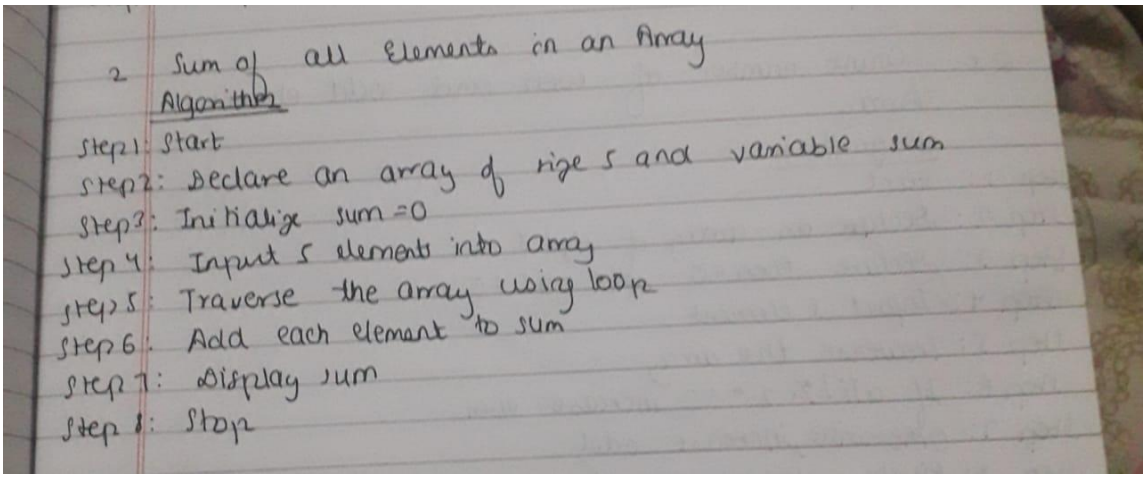
The elements are:
3 3 4 5 6
--------------------------------
Process exited after 22.09 seconds with return value 0
Press any key to continue . . .
```

Time and Space Complexity

- **Time Complexity: O(n)**
- **Space Complexity: O(n)**

Q2. Program to Find the Sum of All Elements in an Array

Algorithm:



Code:

```
#include <stdio.h>
int main()
```

```c
{
    int a[5], i, sum = 0;
    printf("Name: Divya Sharma\n");
    printf("Registration No: 2025-1-53-0323\n");
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &a[i]);
    }
    for(i = 0; i < 5; i++)
    {
        sum = sum + a[i];
    }
    printf("Sum of all elements = %d\n", sum);
    return 0;
}
```

Output:

```
Name: Divya Sharma
Registration No: 2025-1-53-0323

Enter 5 elements:
4
7
2
5
6

Sum of all elements = 24

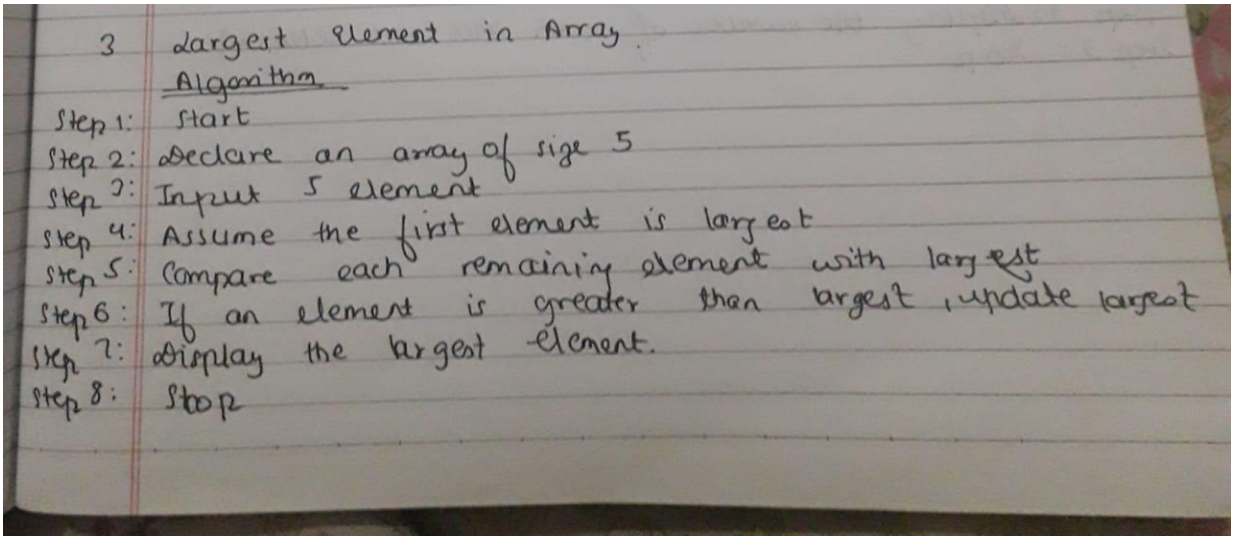
--------------------------------
Process exited after 7.309 seconds with return value 0
Press any key to continue . . .
```

Time and Space Complexity

- **Time Complexity: O(n)**
- **Space Complexity: O(n)**

Q3. Program to Find the Largest Element in an Array

Algorithm:



Code:

```
#include <stdio.h>
int main()
{
    int a[5], i, largest;
    printf("Name: Divya Sharma\n");
    printf("Registration No: 2025-1-53-0323\n");
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &a[i]);
    }
    largest = a[0];
    for(i = 1; i < 5; i++)
    {
        if(a[i] > largest)
        {
            largest = a[i];
        }
    }
    printf("Largest element = %d\n", largest);
    return 0;
}
```

Output:

```
Name: Divya Sharma
Registration No:2025-1-53-0323

Enter 5 elements:
7
1
2
3
5

Largest element = 7

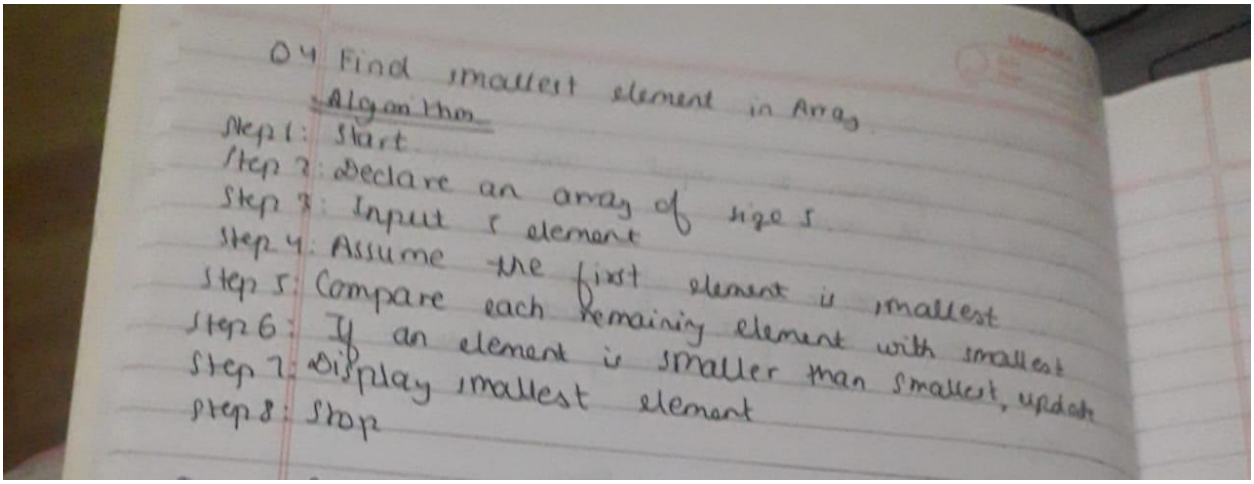
--------------------------------
Process exited after 8.065 seconds with return value 0
Press any key to continue . . .
```

Time and Space Complexity

- Time Complexity: O(n)
- Space Complexity: O(n)

Q4. To find the Smallest Element in an Array

Algorithm:



Code:

```
#include <stdio.h>

int main()
```

```c
{
    int a[5], i, smallest;
    printf("Name: Divya Sharma\n");
    printf("Registration No: 2025-1-53-0323\n");
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &a[i]);
    }
    smallest = a[0];
    for(i = 1; i < 5; i++)
    {
        if(a[i] < smallest)
        {
            smallest = a[i];
        }
    }
    printf("Smallest element = %d\n", smallest);
    return 0;
}
```

Output:

```
Name: Divya Sharma
Registration No:2025-1-53-0323

Enter 5 elements:
2
6
1
0
9

Smallest element = 0

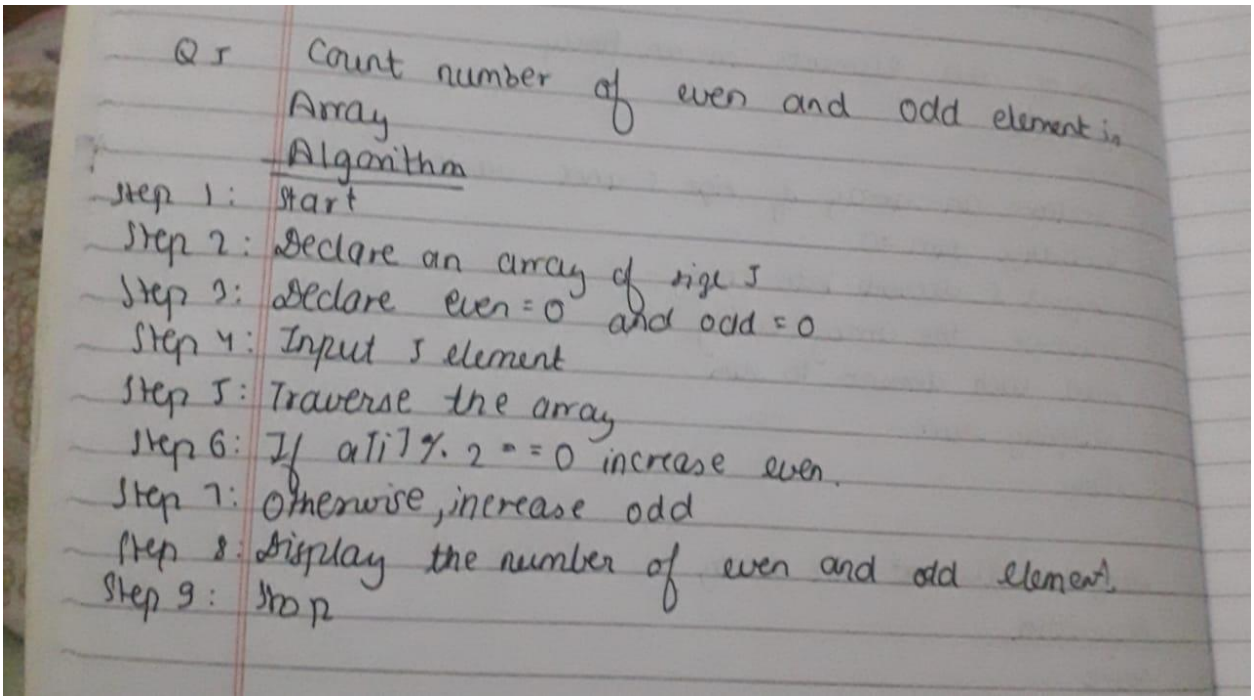
--------------------------------
Process exited after 6.275 seconds with return value 0
Press any key to continue . . .
```

Time and Space Complexity

- **Time Complexity: O(n)**
- **Space Complexity: O(n)**

Q5. To Count Even and Odd Elements in an Array

Algorithm:



Code:

```
#include <stdio.h>
int main()
{
    int a[5], i, even = 0, odd = 0;
    printf("Name: Divya Sharma\n");
    printf("Registration No: 2025-1-53-0323\n");
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &a[i]);
    }
    for(i = 0; i < 5; i++)
    {
        if(a[i] % 2 == 0)
        {
            even++;
        }
        else
        {
            odd++;
        }
```

```
    }
    printf("Number of even elements = %d\n", even);
    printf("Number of odd elements = %d\n", odd);

    return 0;
}
```

Output:

```
Name: Divya Sharma
Registration No: 2025-1-53-0323

Enter 5 elements:
4
3
8
2

0

Number of even elements = 4
Number of odd elements = 1

--------------------------------
Process exited after 9.904 seconds with return value 0
Press any key to continue . . .
```

Time and Space Complexity

- **Time Complexity: O(n)**
- **Space Complexity: O(n)**